

U.S. IT Hardware

Global Memory: New Memory LTAs Part 3 - Bull vs Bear Debate - Webinar slides, transcript and summary



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Following our deep dives into New Memory LTAs (see [Part 1](#) and [Part 2](#)), on July 9th we hosted the webinar New Memory LTAs - glass half full or half empty? This note summarizes the webinar with slides and Q&A. A replay is available [here](#).

LTAs won't remove memory cyclicalities, but they can meaningfully reduce the severity of downturns by raising the earnings floor and improving revenue and profit visibility. LTAs effectiveness ultimately depends on whether the financial guarantees charged are substantial enough to offset the savings customers could realize by purchasing memory at prevailing lower market prices. LTAs can provide meaningful downside protection, but remain vulnerable to abandonment if market prices fall substantially.

While LTAs are unlikely to penetrate into the entire end market, the customers most inclined to enter these agreements - hyperscalers and AI infrastructure providers - are also among the industry's higher-quality counterparties. With strong balance sheets, diversified revenue streams, and strategic incentives to secure long-term supply, these customers may be better positioned to honor their commitments. Micron taking equity stakes of Anthropic is however an example of vendor financing & risk amplification. In addition, unlike prior semi-cycles that were inflated by inventory hoarding or double-ordering, today's demand is more supported by genuine infrastructure deployment for AI.

Although financial guarantees appear modest relative to the massive multi-year revenue streams they are designed to protect, these New Memory LTAs become increasingly protected over time as the guarantees are back-end weighted. As remaining purchase obligations decline, the financial cost of walking away increases relative to contract value. Hence, downside protection can improve materially in the later years of the agreements (when it is needed most!).

On industry fundamentals, we remain constructive on near-to-medium-term memory supply-demand dynamics. We do not expect Chinese entrants to materially disrupt the market over the next two years, with industry supply growth driven primarily by capex decisions at mature incumbents. We also expect HBM pricing to move higher as conventional memory becomes increasingly profitable. Where views diverge is on NAND: Asia team is more cautious, arguing that AI demand disproportionately benefits DRAM and that Chinese competition poses a greater long-term risk in NAND. U.S. team believes AI workloads are becoming increasingly NAND-intensive through larger context windows and growing storage requirements, creating a favorable supply-demand backdrop amid constrained supply.

We highlight a clear divergence in how memory suppliers are approaching LTAs. Sandisk and Micron have been the most proactive in pursuing agreements backed by financial guarantees, while Asia based suppliers have taken a more selective approach. Looking ahead, both teams view further LTA disclosures, memory ASP, and Sandisk's upcoming Analyst Day as key catalysts that could shape investor perception of both LTAs and memory sector valuations.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	17 Jul 2026	Price	TTM	Reported EPS				Reported P/E (x)		
			Closing	Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
SNDK (SanDisk)	O	USD	1,354.82	3,000.00	3092.8%	USD	2.99	65.43	243.73	453.1	20.7	5.6
005930.KS (SEC- Samsung)	O	KRW	253,500	440,000	237.0%	KRW	6,611.53	48,393	77,273	38.3	5.2	3.3
005935.KS (SEC-Pref - Samsung)	O	KRW	172,000	374,000	180.8%	KRW	6,611.53	48,393	77,273	26.0	3.6	2.2
SMSN.LI (Samsung)	O	USD	4,200.00	7,350.00	217.3%	USD	116.15	812.39	1,290.80	36.2	5.2	3.3
000660.KS (SK hynix)	O	KRW	1,830,000	3,300,000	533.2%	KRW	60,341	395,677	568,862	30.3	4.6	3.2
MU (Micron)	O	USD	848.95	1,300.00	623.7%	USD	8.29	67.39	158.99	102.5	12.6	5.3
285A.JP (KIOXIA)	U	JPY	52,110	40,000	2087.2%	JPY	1,014.00	10,013	9,656.84	51.4	5.2	5.4
SPX			7,457.69									
ASIA			1,865.18									
EM			1,732.38									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

SNDK, MU estimate is Adjusted EPS; SNDK, MU valuation is Adjusted P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate **SNDK** Outperform with PT of \$3000.

Samsung Electronics: We rate Samsung Electronics Outperform with price target of KRW 440,000.

SK hynix : We rate SK hynix Outperform with price target of KRW 3,300,000.

Micron: We rate Micron Outperform with price target of US\$1,300.00.

KIOXIA: We rate KIOXIA Underperform with price target of JPY 40,000.

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DETAILS

Following our deep dives into New Memory LTAs (see [Part 1](#) and [Part 2](#)), on July 9th we hosted the webinar New Memory LTAs - glass half full or half empty? This note summarizes the webinar with slides and Q&A. A replay is available [here](#).

The webinar featured Mark Newman, the U.S. IT Hardware analyst and Mark Li, the Asia Semiconductors and Equipment & Global Memory analyst. For Mark Newman's Slides, please see [here](#). For Mark Li's slides, please see [here](#).

PART I: BULL VS BEAR

THE BEAR THESIS: LTA LIMITATIONS AND CONSTRAINTS

LTAs protect future revenue only when the financial penalty for breaking them exceeds the savings a customer could obtain by purchasing memory elsewhere. Customers will generally honor these agreements when the forfeited guarantee or other penalties make walking away uneconomic. However, if spot prices fall far enough below contract prices, the savings may outweigh the penalty, making abandonment financially rational. The strength of an LTA therefore depends not simply on the existence of a contract, but on whether its economic terms remain sufficiently binding under changing market conditions.

The true protective value of LTAs is difficult to quantify. The relevant variables are all changing over time: market prices, contract floor prices, remaining purchase obligations, and the balance of any guarantees. Because investors do not know how these variables evolve throughout the life of each agreement, calculating the exact amount of downside protection becomes more of a conceptual exercise than a precise analytical one.

Not every customer is a natural candidate for a long-term agreement. Hyperscalers and HBM customers have strong incentives to secure supply years in advance, while consumer electronics companies, transactional buyers, and some Chinese customers often prioritize flexibility and lowest-cost sourcing. Even under optimistic adoption assumptions, a meaningful portion of memory demand is therefore likely to remain outside LTAs and exposed to the traditional forces behind the industry's booms and busts.

The amount of protection disclosed so far appears small relative to what it is supposed to protect. Memory suppliers are discussing agreements that span several years and support enormous future revenue streams. Yet the guarantees announced to date represent only a small fraction of that revenue base. The implication is not that LTAs are worthless, but that investors may be overstating how much protection current guarantee levels actually provide. Protecting peak-cycle earnings would require customer commitments on a completely different scale. If the goal is to preserve today's profitability through a severe downturn, the amount of capital that would need to be posted upfront is enormous. Once the numbers are viewed in relation to the industry's multi-year revenue opportunity, the required guarantees start to look measured in hundreds of billions—or potentially even more.

LTAs can soften the cycle, but they cannot replace favorable industry fundamentals. The biggest benefit may simply be reducing the severity of future downturns. Margins that historically might have become deeply negative could instead remain around breakeven. But sustaining elevated earnings for years still requires the same thing the memory industry has always needed: a durable imbalance between supply and demand. LTAs can help at the margins, but they cannot create that imbalance on their own.

EXHIBIT 1: **Differing Guidance Issued by Micron and SanDisk on the Beneficiary of Financial Agreements****On how financial guarantees work, Micron & SanDisk notably guided differently**

- SanDisk suggested financial guarantees to stay as a fixed amount. This implies as RPO lowers over time, the ratio of financial guarantees/RPO rises & presents an increasingly higher hurdle for customers to walk away from LTAs.
- Micron however said the guarantees will be returned to customers on a predefined schedule weighted toward the latter half of LTAs.

	Micron	SanDisk	KIOXIA
Number of Agreements Signed	16 in total as of Jun 2026, including 4 very large and 3 medium sized customers	5 deals, 3 in FQ3, 2 so far in FQ4	
Total RPO at Min Price (US\$B)	100B for 14 of the deals	42B for 3 deals	
Contract Length	Typically 5 years from CY26 to CY30. Generally 3 years for auto customers	Up to 5 years from FY27	Multi year starting CY27
Coverage of LTA	When completed, half or more revenue is expected to be under LTA. In total, the 16 LTAs cover 20% of Micron's DRAM volume and 1/3 of NAND volumes in 5 years.	5 deals will account for 1/3 of SNDK's bit demand in FY27	Plan to cover ~50% of CY28 shipment
Pricing Commitments	Largest LTAs generally have price <u>ceilings at CQ2 market price</u> , and a <u>floor price that guarantee margins well above past peaks</u>	A combination of fixed and variable pricing	
Financial Guarantee (US\$B)	22B	11B	
Return of Financial Guarantee	Financial guarantee <u>will be returned to customers on a predefined schedule weighted towards latter half of the LTAs</u>		Financial guarantee is likely <u>kept unchanged through out the life of LTAs</u>

Note: RPO = Remaining Performance Obligations
Source: Company reports and Bernstein analysis

Source: Company reports and Bernstein analysis

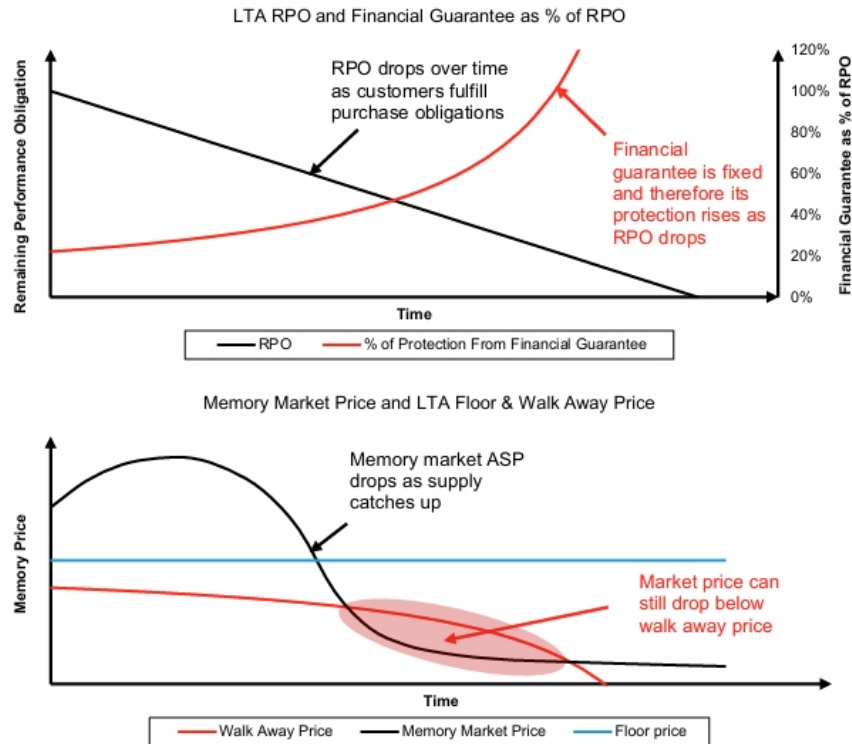
EXHIBIT 2: **Contract Details & Memory Prices Over Time Determine LTA Effectiveness****The effectiveness of LTA depends on contract details & price over time & is impossible to assess in a bottom-up manner**

- When trigger customers will walk away from LTA
 - $\text{RPO} - \text{market price} * \text{remaining volume} > \text{remaining financial guarantees}$, or
 - $(\text{floor price} - \text{market price}) * \text{remaining volume} > \text{remaining financial guarantees}$
- So as long as remaining volume > 0 , it is mathematically possible to walk away if market price is low enough vs. floor.
- The effectiveness of LTA = integral of
remaining financial guarantees – [(floor – market price) * remaining volume]
over time & is impossible to quantify without details.

Source: Bernstein analysis

EXHIBIT 3: **There are Memory Market Conditions Where Customers can Benefit by Backing Out of Their LTAs**

Mathematically falling memory price can still trigger customers to walk away from LTAs, even if protection from financial guarantee gets stronger over time

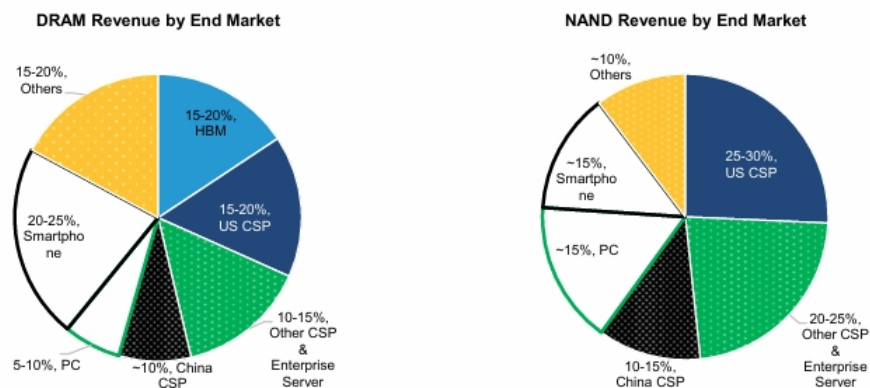


Source: Company reports, Bernstein estimates and analysis

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 4: **LTA Penetration Has a Limited Ceiling****How much of memory business can be covered by LTA eventually as more negotiations conclude?**

- We understand negotiations with US CSPs are mostly done, & still ongoing with Chinese CSPs. Enterprise & consumer customers will follow.
- In a recent [event](#) SanDisk CFO said consumer business is “more transactional”, and these LTAs “don’t really apply.”
- Chinese customers should prefer signing LTAs with Chinese suppliers who will take away an increasing part of the market.
- So perhaps 30-50% of the market can never be covered by LTA?

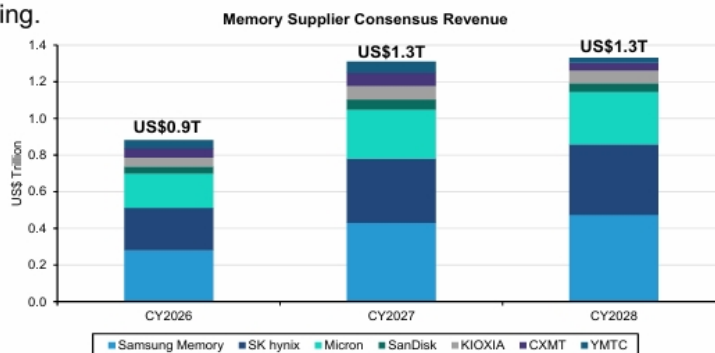


Source: Company reports, Gartner, TrendForce, Bernstein estimates and analysis

Source: Company reports, Gartner, TrendForce, Bernstein estimates and analysis

EXHIBIT 5: **Current LTA Deposit Levels Appear Insufficient to Structurally Reduce Earnings Volatility****Top-down, with deposits <1% of the revenue needed to protect so far, can LTA be enough to avoid earnings volatility**

- Consensus expects memory companies to have US\$1.3T revenue/year in CY27 & 28.
- If LTA can keep this yearly revenue for 3-5 years, LTA needs to protect c. US\$5.2T.
- So far the guarantees is only US\$33B, i.e. 0.6% of the protected amount.
- Samsung & SK hynix will likely secure more guarantees, but that unlikely will make a difference.
- Even 20% of the protected amount is US\$1T. Who on earth can do such as a massive deposit upfront?
 - The MU-Anthropic deal underscores the need of memory but is achieved with an equity stake – i.e. vendor financing.



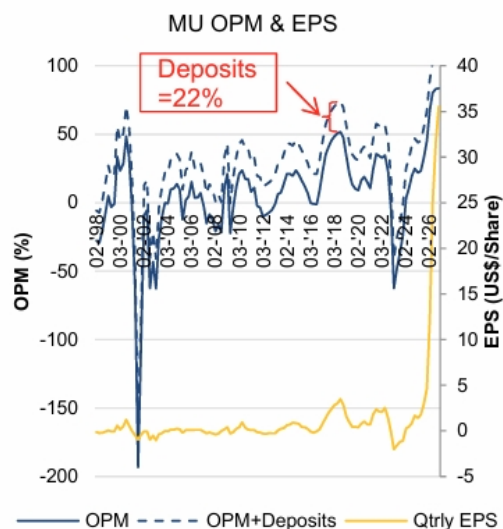
Note: CXMT & YMTC revenue is based on Bernstein estimates

Source: Company reports, Bloomberg, Bernstein estimates and analysis

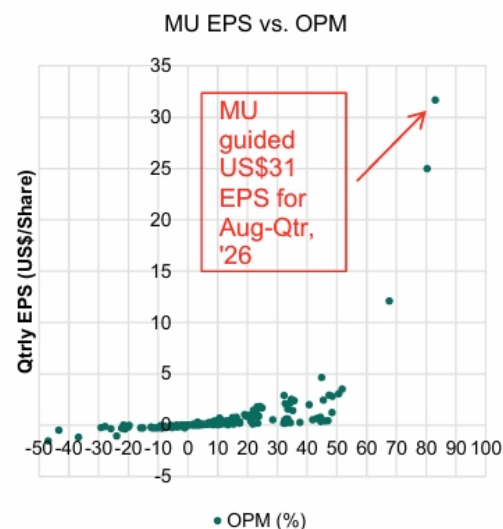
Source: Company reports, Bloomberg, Bernstein estimates and analysis

EXHIBIT 6: **Guarantees Can Lift Trough Economics but Cannot Eliminate Memory Earnings Cyclicity****Financial guarantees makes downturns more palatable & motivates capacity expansion, but it can't keep earnings at a very peaky level**

- Deposits given up by customers should make trough margins more palatable, say OPM 0-30% (up from -10 to 10%) if we ignore the 2 worst cases in history of Micron.
- If OPM is 0-30%, then EPS would be only 1/10 of the current EPS run rate.



Source: Company reports, Bloomberg, Bernstein estimates and analysis



Source: Company reports, Bloomberg, Bernstein estimates and analysis

EXHIBIT 7: **Bear Case Summary**

Summary

- To make LTA have teeth, advanced cash deposit is critical, but we think no one can come up with such a massive upfront deposit to sustain the peaky earnings for 3-5 years, probably not even for a year.
- LTA helps mitigate the downside but can't sustain earnings.
 - Good for semicap, but not enough for memory stocks.
- It takes continued shortage to sustain earnings. I.e. for the earnings to be structural, the shortage needs to be structural.

Source: Bernstein analysis

THE BULL THESIS: THE NEW MEMORY LTA'S BENEFITS AND PROTECTIONS

Today's memory LTAs are fundamentally different from many of the failed contract structures investors point to. Past examples often lacked meaningful financial guarantee. In some cases customers could walk away with little consequence; in others, suppliers technically had contractual protection but had to spend years in court trying to recover losses. Those experiences understandably made investors skeptical, but the current agreements are structured quite differently.

The most important difference is the presence of upfront financial commitments. Customers are not simply promising future purchases. They are posting deposits and guarantees that have immediate economic value. That creates a tangible cost to breaking the agreement and gives suppliers protection that many earlier semiconductor contracts never had.

The structure of the guarantees becomes stronger as time passes. A key feature of these agreements is that financial commitments are back-end weighted. As the remaining purchase obligation declines, the relative protection provided by the guarantee increases. By the later years of a contract, a customer may be giving up a significant amount of value to walk away from a much smaller remaining commitment.

The quality of the counterparties is another important distinction. Many of these contracts are being signed with hyperscalers and other large technology companies that have exceptionally strong balance sheets. Unlike some historical examples where customers were financially stressed, these counterparties are viewed as highly creditworthy and accountable. That makes the contracts more credible from the outset, and even if there is any shortfall beyond posted collateral, it is realistically collectible.

The demand backdrop also looks fundamentally different from prior industry examples. In several past cases, long-term agreements unraveled after demand turned out to be artificially inflated by inventory hoarding, double-ordering, or pandemic-era supply chain behavior. However, the current memory demand is being driven by genuine consumption rather than temporary distortions. Moreover, the demand is tied to AI infrastructure and rapidly growing training/inferencing requirements, which means memory has exponential scaling potentials.

Even under very harsh assumptions, LTAs appear capable of providing meaningful downside protection. Our stress tests assume severe memory price declines and, importantly, allow customers to walk away whenever doing so makes economic sense. Despite those assumptions, earnings remain materially better than they would in a world without LTAs. The contracts do not eliminate downside, but they do appear to reduce it significantly, especially in the outer years.

The analysis may actually understate the real-world effectiveness of these agreements. The models generally assume customers behave with perfect economic rationality and ignore most non-financial considerations. In reality, a customer may think about litigation risk, reputational damage, future access to supply, and the importance of maintaining strategic supplier relationships. Those factors could meaningfully raise the threshold for walking away.

None of this means memory cyclicity disappears, but it will be meaningfully dampened. Prices can still fall, earnings can still decline, and contracts can still be broken under the right circumstances. The claim is not that LTAs permanently fix the memory industry. Rather, they change the magnitude of the cycle by making it harder for revenue and earnings to collapse in the way they have historically.

The most balanced conclusion is that LTAs raise the floor and offer robust protection during down cycles. They should not be viewed as a mechanism for locking in peak earnings forever. Instead, their value lies in reducing volatility, increasing visibility, and providing a meaningful buffer against future downturns. The memory business remains cyclical, but potentially less cyclical than it has been in the past.

For additional details on LTA stress tests, please see our [Memory: What do New Memory LTAs mean for sustainability of earnings and multiples? Raising SNDK TP to \\$3000](#).

For additional details on why we believe the New Memory LTA is different, please see our [Memory: New Memory LTAs part 2 - Case studies from Microchip and other Semiconductor LTAs - What's different this time?](#).

EXHIBIT 8: **Key Messages****Key Messages**

- **Historically, long-term agreements in semiconductors and memory functioned primarily as a buyer's tool.** Customers used LTAs to secure guaranteed volume, while sellers had little structural leverage of their own.
- **Both structures of historical LTA contracts offered sellers little real downside protection, since neither carried an upfront financial commitment** to compensate the supplier if a buyer walked away.
- **Two case studies illustrate exactly how previous LTA structures failed in practice:** 1. Microchip / Analog Semiconductors; and 2. Hemlock / Polysilicon. Hemlock case bears more resemblance to what we see in the New Memory LTAs.
- **We identify three structural differences between legacy semiconductor LTAs and the New Memory LTAs.**
 1. **Upfront financial commitment:** back-end-weighted cash deposits, letters of credit, or third-party guarantees)
 2. **Better counterparty quality:** financially sound, diversified buyers replace the thinly capitalized or diffuse customer bases that undermined Hemlock and Microchip
 3. **Structural demand:** sustained AI-driven buildout replaces the temporary pull-in and double-ordering that ultimately undermined Microchip(sustained AI-driven buildout replaces the temporary pull-in and double-ordering that ultimately undermined Microchip)
- **The New Memory LTAs provide meaningful but not infinite downside protection.** We find that these new memory LTAs are structured in a way which provides significantly higher disincentives to customers to walk away - particularly in the latter part of the contract term - than the cases we previously analyzed in our first note.

Source: Bernstein analysis

EXHIBIT 9: **New Memory LTA Contracts Protect Vendors More than Their Predecessors****New memory LTA contracts are designed structurally differently from their predecessors.**

Exhibit 1 - Old LTAs vs New Memory LTAs: this time it is really different

Dimension	Microchip PSP	Hemlock polysilicon	SNDK / MU new memory LTAs
Industry structure	Fragmented; dozens of substitutable analog/MCU suppliers with no individual pricing power	Moderately concentrated among a few global producers	Consolidated oligopoly (roughly 3 DRAM producers / 6 NAND makers)
Demand growth profile	Shortage-driven spike plus double-ordering that reversed into a glut once customers over-inventoried	Policy/subsidy-driven solar boom-bust; genuine shortage flipped into classic commodity oversupply after 2008	Structural AI/data-center buildout; high-end memory sold out through 2026 into 2027, with both sides planning into 2027-30
Customer concentration	Extremely diffuse; PSP backlog spread across a huge base	Small solar module makers, several of which later failed or entered insolvency	Concentrated among a few high-quality hyperscalers/AI buyers
Customer type / balance sheet	Mixed industrial/auto OEMs and distributors; varied, often mid-size credit quality	Thin-margin, subsidy-dependent solar developers with weak balance sheets; SolarWorld AG filed for insolvency	Fortress-balance-sheet mega-caps able to post \$10-20B+ of cash and letters of credit
Customer unit economics / incentive to honor	End-demand softened; carrying excess inventory cost more than walking away, weak incentive to honor NCNR terms	Honoring meant paying above-market feedstock prices while competitors bought cheaper spot material, strong incentive to breach and litigate	AI infrastructure is business-critical and memory gates a trillion-dollar opportunity; the deposit is trivial vs. the cost of being short, strong incentive to honor
Contract terms / enforceability	12-month rolling NCNR backlog, no cash; relaxed to 6 months in Aug 2023, then discontinued for new orders in Feb 2024, reputational, not enforced	Fixed-price take-or-pay; enforceable only via years of litigation, and even a ~\$793M win proved largely uncollectible against an insolvent counterparty	Multi-year volume and price commitments backed by upfront cash deposits, letters of credit, or third-party guarantees held by (or accessible to) the supplier, with guarantee's coverage ratio rising toward 75-100% of remaining obligations in the outer years, offering real downside protection

PSP = Preferred Supplier Program; NCNR = Non Cancelable, Non Reschedulable
Source: Company reports, Bernstein analysis

Source: Bernstein analysis

EXHIBIT 10: **Downside Protection for Vendors is Baked Into New Memory LTAs****Downside protection is ingrained into contracts through back-end-weighted deposits.**

Exhibit 2 - Paper claim vs Real Protection

	Hemlock Take-or-Pay	New Memory LTAs (SNDK/MU)
Financial Guarantee at signing	No significant long-term upfront financial guarantee that is backend-weighted like SNDK/MU, only prepayments in installments credited towards future purchase	Cash deposits / letters of credit / third-party guarantees posted upfront, returned towards the back of the contract
Nature of protection	Legal claim; supplier must sue to collect	Financial guarantee already held by (or accessible to) the supplier
Remedy if buyer walks	File suit, litigate for years, then attempt to collect	Draw on financial guarantee directly
Timing of protection	Static; contract terms don't strengthen over time	Dynamic; guarantee covers a shrinking remaining obligation as revenue is recognized, so the guarantee's coverage ratio is higher over time
Strength in outer years	Weak - Hemlock only had some prepayments, no financial guarantee that is weighted towards the backend of the contracts	Strongest - back-loaded return schedule means coverage ratio rises toward 75-100% of remaining RPO in the outer years
Outcome when buyer distressed	Judgment often uncollectible (SolarWorld AG insolvency left Hemlock exposed even after winning)	Financial Guarantee already sits with the supplier or a trusted third party, so buyer insolvency doesn't erase the protection
Result	A won lawsuit that provided little real economic recovery	A structurally strengthening cushion that protects margins and earnings precisely when a downcycle bites

Source: Company reports, Bernstein analysis

Source: Bernstein analysis

EXHIBIT 11: **Contract Counterparties in New Memory LTAs are Healthier than Before****New memory LTAs involve healthier, more robust counterparties.**

Exhibit 3 - New Memory LTAs: Better Counterparty Risk Profile

	Hemlock (e.g. Sachsen)	New memory LTAs (e.g. hyperscalers)
Type of customer	Solar wafer and module manufacturers	Hyperscalers, Large OEMs, and AI Infrastructure providers
Customer revenue structure	Economics depended almost entirely on the solar value chain; polysilicon was a single, dominant point of exposure	Diversified across cloud infrastructure, advertising, software, and enterprise services; memory is one input among many
Impact of honoring the contract for customer	Take-or-pay price diverging from a collapsed spot market was large enough to destroy margins and the business itself	Memory cost is a small fraction of overall earnings, absorbed without threatening core business survival
Number of customers	Numerous, dispersed solar manufacturers	A handful of hyperscalers
Financial strength of customers	Several thinly capitalized, frequently under financial distress	Investment-grade balance sheets, transparent capital plans
Collectability if shortfall occurs	Collection depended on solvency of a distressed customer base, even after winning in court	Accountable and reliable; shortfalls beyond posted collateral are realistically collectible

Source: Company reports, Bernstein analysis

Source: Bernstein analysis

EXHIBIT 12: **Memory Demand Is Structurally Expanding in the AI Era****Memory demand is structurally higher than the past due to growing AI needs.**

Exhibit 4 - Analog vs Memory demand: Physical Limits vs. Exponential AI Scaling

	Analog / Microcontroller Chips	Memory Chips (DRAM/NAND/HBM)
Nature of demand	Essential but finite - like oxygen, once you have enough, you don't need a molecule more	Foundational to AI capability - more memory directly makes AI models smarter and more capable
Scaling behavior	Does not scale indefinitely; demand is a fixed allocation tied to unit production (cars, appliances, industrial equipment)	Scales continuously with AI workload growth - larger models, more inference, more data centers all require more memory
Driver of demand during shortage	Mostly a pull-forward: customers over-ordered during Covid shortage to build safety stock, not because true consumption increased	Structural buildout: driven by genuine, compounding growth in AI training and inference workloads
Underlying economic logic	Fixed ration to be filled once, then forgotten	Continuously expanding resource requirement tied to AI capability growth
Result for LTA credibility	"Non-cancellable" orders proved unenforceable in practice once the pull-forward reversed	Long-term commitments backed by real collateral are credible because underlying consumption is durable and strategically irreplaceable

Source: Company reports, Bernstein analysis

Source: Bernstein analysis

EXHIBIT 13: Memory LTAs Are Driving Greater Margin and Pricing Stability

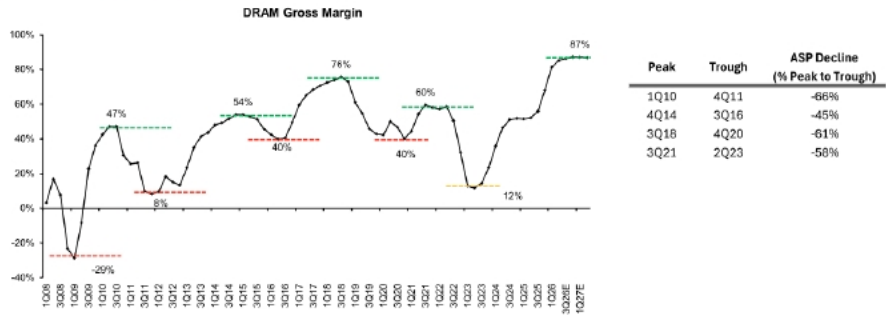
Historical NAND and DRAM downturns

Exhibit 6 - NAND peak to trough ASP decline



Source: Trendforce, Bernstein estimates and analysis

Exhibit 7 - DRAM peak to trough ASP decline



Source: DRAMeXchange, Bernstein estimates and analysis

Source: Bernstein analysis

EXHIBIT 14: Implied SanDisk EPS Robust Under Various Penetration and Market ASP Regimes

SNDK EPS: New Memory LTAs provide significant (but not infinite) downside protection.

Exhibit 16 - FY29 - Implied EPS

FY29 - Implied EPS					
LTA Penetration	\$0.23	\$0.17	\$0.11	\$0.06	Market ASP
	-44%	-58%	-72%	-86%	Vs. Bern Peak ASP of \$0.41
0%	\$ 186.70	\$ 125.12	\$ 63.54	\$ 1.95	
20%	\$ 199.02	\$ 149.75	\$ 96.10	\$ 34.52	
40%	\$ 211.33	\$ 174.39	\$ 128.67	\$ 67.09	
60%	\$ 223.65	\$ 199.02	\$ 161.24	\$ 99.66	
80%	\$ 235.97	\$ 223.65	\$ 193.81	\$ 132.22	

Source: Company reports, Bernstein estimates and analysis

Exhibit 17 - FY30 - Implied EPS

FY30 - Implied EPS					
LTA Penetration	\$0.23	\$0.17	\$0.11	\$0.06	Market ASP
	-44%	-58%	-72%	-86%	Vs. Bern Peak ASP of \$0.41
0%	\$ 228.73	\$ 154.91	\$ 81.08	\$ 7.25	
20%	\$ 243.50	\$ 184.44	\$ 125.38	\$ 66.31	
40%	\$ 258.26	\$ 213.97	\$ 169.67	\$ 125.38	
60%	\$ 273.03	\$ 243.50	\$ 213.97	\$ 184.44	
80%	\$ 287.79	\$ 273.03	\$ 258.26	\$ 243.50	

Source: Company reports, Bernstein estimates and analysis

Source: Bernstein estimates and analysis

PART II: Q&A**Q = Michael Tedesco, MN = Mark Newman, ML = Mark Li**

Q: We already can see some differences in terms of how you're viewing the LTAs. Why do the LTAs even matter for these stocks? Because if they actually have to enforce them, either by shoving volume down at their customers or taking hostage of deposits, these stocks are all going to sell off anyway, was the viewpoint of at least one of our clients. And if you're right about this, Mark Newman, the LTAs being a buffer, a protection, then when are we gonna actually see multiples expand to 10 times plus?

MN: Yeah, so the question is, if price collapses, who cares about LTAs because it's a down cycle? I mean, I get that. In any down cycle, earnings are going to get pressured. I think that's pretty clear from the sensitivity analysis. They're not going to sign 100% on the LTA protection. That's not on the cards. But we think they're going to go above 50%. That was what I think was stated by Micron and by Sandisk that they're going for over 50%. But it won't be 100%, so there's going to be downside protection, but I think once investors get a little bit more comfort and understanding that it does reduce downside risk, then it can help the multiples. I do think, though, what is also important, which is difficult to predict, is how fast is the downturn and when? Because if the downturn was rapid and now, then all bets are off because there's much less protection in the earlier years because of the nature of the financial commitment. It's much, much more back-end loaded, much, much more protection in the back end. So, if you make a very bearish assumption that demand collapses in very near term, then it's worse than what I was projecting here. And so, I think that stocks would be at more risk there. But my point is, looking at longer term, we think that prices aren't gonna collapse in the short to medium term. It's maybe more risk 2 years out, and so you have much more protection going out that far.

Q: Okay. Mark Li, thoughts?

ML: I think LTA can, say, make the trough a bit better, or not as bad. But there will be still correction. There will still be peak or trough. But LTA can make the trough, yeah not as bad, especially the LTA that comes with the deposits.

Q: And I think your argument would be that these stocks don't deserve to trade it more than 10 times. So that part of the question is probably not applicable to you, correct?

ML: Yeah, for peak earnings. Earnings change every quarter, meaningfully, you can say. So yeah, 5 times on peak earnings or 5 times on, say, sustaining earnings is very different.

Q: Let's start with the question that got the most votes, which was that Micron just raised their investment commitment from \$200 billion to \$250 through 2035. Also agreed to some investments in the US ecosystem at an additional \$30 billion. Isn't this essentially an upstream LTA for domestic wafer suppliers? And who benefits the most? I think, Mark Li, that's probably one for you.

ML: It's not as binding an LTA, it's just an announcement. I don't think it's a contract per se. It's not as binding, I would say. It's, 10 years from now. So with that, things can change. So, frankly, I will not read that much into the announcement. But if anything, yes, good for semicap. Especially good for, you can say for builders who build construction in the US or Micron.

MN: If I could just add, because I think this is something that we both agree on, it's that demand is very strong for memory. Memory companies are gonna have to increase CapEx. Fab costs are getting more and more expensive. This is applicable to both DRAM and NAND. And so, these big numbers seem big and scary, but there definitely needs to be more CapEx considering how much shortage we're in now.

Q: Can you talk about China? And will Chinese supply disrupt the cycle? And is the technology good enough to matter right now?

ML: At current level or the current size, which is small right now, China in both DRAM and NAND are unlikely to change the shortage meaningfully in the next two years. Let's focus on DRAM first. In DRAM China's share is only 8% according to IPO prospectus of CXMT in 1Q 26. We are expecting China will be, say, nearly doubling their global share to 16% in about maybe 3-4 years. But they only add 8% supply incrementally into the market. But imbalance right now, I believe it's probably 20%, 30%, maybe, imbalance. Samsung even already mentioned 30% to 50% in a recent webinar. And the IR refused to offer detail when we pressed them for details. But the size of imbalance is more meaningful than what the addition by China can really change. So our view is China could be a long-term threat, especially in NAND, given you don't need EUV to expand NAND capacity, and to make NAND better. But in terms of supply-demand, you still count on incumbents to maintain the imbalance, and therefore, to keep earning at peak levels. Maybe they can do that, maybe they can make the imbalance small enough in 2028. Then China can be a swing factor. But by 2028, it would likely not happen. Definitely not this year, and definitely not in 2027.

Q: Okay, Mark Newman, any thoughts there?

MN: Yeah, I mean, there's nothing I disagree with. I think I totally agree with that. I think the risk is more longer term from China, and it's in all of the supply-demand models. China is in there, the CXMT and YMTC, it's in there. But as Mark Li said, the bigger impact to supply demand is going to be the really huge suppliers, Samsung, Hynix, Micron. What they do. They've been very constrained for a while in both DRAM and NAND. And so, as time goes on, they are all building new cleanroom space. And so by 2028, they have more ability to add capacity. So I think that's the more meaningful risk to supply-demand. And then I think China more of a structural concern, looking out longer term. If you have an added player in the market that is becoming bigger and more competitive, it kind of just gets more concerns longer term around, say, sustainability of earnings and the cycle and all things like that longer term. But at the moment, it's really about how strong is AI demand and supply additions from the others.

Q: Got it. Got it. Thank you. Any thoughts on the contracts or LTAs for high bandwidth memory and how HBM price changes over the next year and market share?

ML: The contract discussion is ongoing. So far, HBM price hasn't changed because the price has been locked-in in year 2026 contracts. Now, suppliers and customers are discussing 2027 contract. And we believe, suppliers are giving pressure to hike prices because now capacity can get much more profit, much more revenue for suppliers if capacity is used to produce conventional memory. So we foresee there will be some concessions made by customers. We assume HBM price will be higher by 2 times to 2.5 times from this year to 2027. They already account for the strategic value of HBM because the price hike in conventional is far bigger. Commercial price has hiked already 4 times, and it could be even more in the next 12 months. So it

could be 5 times from 2026 to 2027. So, we assume the price hike in HBM will be, say, smaller. And therefore the strategic value of HBM is recognized by suppliers. This also means HBM is somewhat volatile, too. We do see HBM a steadier business. But because they share the same capacity with conventional DRAM, so the shortage effect, the volatility of conventional is going to spill over to HBM. It will still be meaningful swing - 2-time swing is still very meaningful - although it's not as volatile as a 4 or 5-time swing as we see in conventional.

MN: Yeah, I agree. I mean, I think it's remarkable that price has gone up so much in conventional, but not in HBM. So it's much less profitable to make HBM than regular DRAM right now, so price has to move. I think Mark Li is completely right there.

Q: Let's shift to NAND. And I think you guys have a little differing viewpoint on NAND. One of the questions here is, why did Micron guide to the NAND cycle ending sooner than the DRAM cycle? So we can start with that. But I think, in general, I'd like to hear your viewpoints on NAND. Obviously, Chinese, supply is more of an issue with NAND than DRAM, or it sounds. But, Mark Li, why don't you start, and then we'll have Mark Newman finish that up.

ML: Okay. Yeah, there are a lot of debates on, say, how much demand AI can bring to DRAM versus NAND. I'm much more constructive on DRAM, the demand for DRAM, less constructive on NAND. But frankly, it's difficult to prove given AI is still evolving. Without detail, we can't quantify precisely. In terms of why Micron said so, I was surprised when I heard that, to be honest. Because Micron had been saying that the investment in NAND would be far smaller. Most of the investment would be DRAM. And it's more than Micron. Samsung, SK Hynix, said the same thing as well. So, when Micron said it, I think they implied the demand for NAND is even lower despite supply addition is lower. So demand is even lower than that, and that's why the shortage in NAND will end sooner.

Q: Mark Newman, differing viewpoint, I believe.

MN: Yeah, I mean, so I think NAND, as you go into different stages of AI-driven demand, I think the earliest stage of AI training and basic chatbot inference was mostly HBM and DRAM intensive. But as we now get into the later stage of AI, where we have more advanced inference, longer context windows, and also moving towards Agentic, you're getting much, much more longer context that needs to be saved. And so in order to make the models smarter, you want to have context of what you searched before. And there's this KV cache or context memory. It sits predominantly in HBM and DRAM, but it is overflowing now into NAND flash, which is why you look at the Vera Rubin spec. They essentially 5x the NAND content in Vera Rubin from 4 terabyte per GPU to up to 20-21 terabyte per GPU. So a significant, significant increase. And I think that's because of the changes you see in the development of AI.

The other thing I'll say about NAND is it's relatively earlier cycle than DRAM because of what we just talked about — the AI helping DRAM first, the demand rebound and price rebound started much earlier in DRAM. DRAM gross margins were already 60% or so the industry level this time last year, whereas NAND was loss making at this time last year. And so NAND prices only started recovering around August. It was a much, much later recovery, partly because AI didn't benefit NAND so much in the early innings, but now it's starting to more. Price increases hadn't been as much. They're now catching up because NAND has recently been increasing faster than DRAM, so it's actually narrowing the gap. But I think that the CapEx response in NAND is delayed versus demand because the DRAM rebound happened earlier and so there's been a lot more focus from the diversified memory makers like Samsung, Hynix, Micron in building new clean room space for DRAM and not for NAND. So it's at the expense of NAND supply. Therefore, supply is much more constrained for NAND until 2028. I think NAND still has a way to go higher. And, I still like Sandisk here. I can talk about that more if there's interest.

Q: When are memory prices going to cause demand destruction now that they're so much of the bomb for AI? And that's a broader question than just AI, but I'll ask you, Mark Newman, to start with that.

MN: Well, I mean, you can already see that memory prices are destroying demand. In consumer products. Smartphone and PC unit growth has been extremely weak. Partly because, the OEMs are having to hike prices and that's destroying demand of these products, PCs and smartphones. They're projected to be down something like low to mid-teens down this year, yoy, for both smartphones and PCs. So we've already seen it there. I think the question is also alluding to when will we see the impact on AI from higher prices? And we haven't seen it yet, but, I think we are seeing, certainly, distress from the customers. Particularly, potentially the de-specking of AI servers because they can't get enough DRAM in them. So, I think it's starting to be a sign of distress, but it's more supply-driven than no demand. There's actually pent-up demand, I think, in the big hyperscalers. They want to add more compute, there is constraint in compute. You see Anthropic is compute constrained, so they want more compute, which means they need more memory and they just can't get it. So, I don't see that demand has been impacted right now, it's just that the supply is not enough to keep up. I don't see that changing anytime soon, but it's very difficult to see. I think it's very important that the hyperscalers get ROI on their investments. And I think that's the worry that we all have, looking at a

year or two years from now, are they going to keep spending at these rates? And it's very difficult to answer that question now.

Q: Mark Li, do you agree with that response?

ML: Yes, but I can offer a bit more details. We did a study using Vera Rubin as example. So we find out if a data center deployed with Vera Rubin. Before the memory price hike, assuming the investment is 1. Now the investment will become 1.3 with the higher memory cost. So "ROI" suddenly will be lower, now with the "I" becoming 1.3 instead of 1. But my view is there's a competition pressure among OpenAI, Anthropic, Google, etc. They are all now still well-funded. So I do think they will continue to invest. So maybe the end result is they'll do some small changes, optimization here and there to make "I" become 1.25. So it's not 1.3, it's 1.25, but it'll be still higher than 1.

Q: There are two questions I think that are very similar, so I'm gonna ask them both. One is, about Hynix Management talking about LTAs and volume commitments with no price floor/ceiling. So, traditional, more traditional, and Micron talking about a price floor. What does this mean for your view and their views on pricing and the sustainability? And on top of that, just maybe talk about the financial guarantees and how they differ between Samsung, Hynix, and Micron. Mark Li, this is more for you, but, I'm sure Mark Newman would want to chime in at the end.

ML: Okay, in the supply chain, what we learned was that it's indeed Sandisk followed by Micron being most active in LTA discussion. The Korean guys are in between. And Kioxia is sort of, the most lukewarm in LTAs. That's what we learned in the supply chain. Now I guess we get some confirmation from SK Hynix. I learned this from a few investors who met the company in their ADR road show too. I guess they are very confident in supply-demand. And therefore, not having a price band, allowing them to get your upside, and the upside is more than downside. Yeah, I believe it could be the case. Or, they don't believe... you know LTA is just a piece of paper... They don't believe any customer can come up with sufficient financial guarantee up front anyway. Yeah, so don't bother to go through all this legal hassle to sign your LTA. Basically, what TSMC did in the past, they never signed an LTA. Because of the same reasons.

Q: Mark Newman, anything to add?

MN: Yeah, I mean, I think just a bit of helpful context, hopefully, is that I think Sandisk has been the one that has been more on the front foot here with these new LTAs, and I think the reason for that is partly just the background of David Geckler, the CEO of Sandisk. He is not a typical memory CEO. He doesn't come from a long-term memory background, unlike most of the other memory companies. And so, he has a background, it's HDD, Cisco, and software. And so he thinks much more out of the box and applies lessons learned from other industries. And so I think, they've been the most front-footed on this, and I think Micron, being a US company has been the next most aggressive, and they've similarly announced quite a lot of innovative new kind of contract structures — fairly similar to what, not exactly the same, but fairly similar to what Sandisk announced. The others are just way behind and not clear to me if they're going to follow suit. I wouldn't be surprised if they do eventually follow suit. But not yet. At least we don't have much clarity that they have done so yet.

Q: This kind of leads to another question that is kind of obvious, is if Sandisk believes that prices are going to continue to rise, why are they pushing customers to put so much aside in escrow?

MN: Well... I think that's fair. It's only, they say, just over 1/3 of volumes in long term agreements currently. But as time goes on and prices presumably continue to go up, at least in the short term, that long-term agreement portion will get higher. But presumably the ceiling is also going up in those agreements. And I think that the company just thought that, rather than having gross margins potentially with more upside risk in the short term, but then no downside protection, they've decided to cap some of the upside in exchange for reducing downside risk, particularly in the outer years. And so, I think from an investor perspective, they should deserve a higher multiple by doing that. That's that's been why they've done it. But, yeah, of course, they're potentially leaving some money on the table. But it's only a third of volume currently.

Q: I am interested to understand how you can have a buy on Sandisk and an underperform on Kioxia. Surely the dynamics are the same. I got something offline saying, how can one trade it? We argue one should trade it 11 times forward and 14 on through cycle PE, while the other only would be worth 4 times. Maybe you can highlight your views in a constructive manner for us.

MN: Well, I think I outlined my high level view on NAND and Sandisk. I mean, basically, NAND is also benefiting from this huge growth in demand. It's earlier cycle than DRAM, meaning the impact of demand has happened later. So the CapEx response has also been later. Therefore, you're going to see supply response potentially hit later than DRAM. I'm not saying I'm negative DRAM at all. I'm just saying that there's, I think, a lot of positive things to think about for NAND flash. And there is clearly a significant

tightness in the market now based on how high prices are going. And, I think right now it is trading around 5 or 6 times our FY28 earnings, and if you look at our bull case, it's trading at even less than that on our bull case. And I think our earnings projections are relatively conservative.

And I think the near-term catalyst for Sandisk, just going to the stock specifically, the near term catalyst is they are going to report, I think it's right at the end of this month. And, I think it's going to be another strong quarter where consensus number estimates are going to be revised up significantly. And they're going to follow that up with, August 13th, they've got an analyst day in New York where they've said they're going to give a lot more clarity on these long term agreements. More clarity in the long-term agreements will be taken positively because it starts to get investors confident that the downside risk is somewhat protected, somewhat muted compared to some of the fears that are out there. So I think those are the two near-term catalysts over the next 5-6 weeks or so that will help drive the share price up. And I think it's cheap where it is now, particularly after the recent sell-off.

Q: Okay. Mark Li on Kioxia.

ML: Yeah, the rating on Kioxia is more on, say, long-term consideration, so it's a more longer-term rating. Mostly because of the threat from China. And also there's sudden demand driven by AI. We share the same view as Micron's guidance implied. Yeah, we see a demand, better demand outlook for DRAM, less so for NAND. Yeah, so these are key reasons. In terms of valuation, to be precise, we use sum of the parts to value Kioxia. We divide earning into a few parts. The portion protected by LTA, we put 20x 1 year forward PE. For the unprotected portion, we use 10x. And we account for the deposit given up by customer. We also account for case return... the company will return to investor, but there will be only 1x, because it's cash. So with that sum of the parts, we find out roughly the equivalent 1 year forward PE is about 5x -ish, roughly. So that's why. I think another reason, if structurally DRAM is better than NAND, then Kioxia has all the reasons to have a lower multiple than Samsung, Micron, and SK Hynix. I think SK and Micron now have also about 5x forward PE. Yeah, I think Kioxia should only have 4x.

Q: I mean, obviously the stock has sold off a little bit in the last month or so, 20-25%, so maybe more than a little bit, but it's had a phenomenal run. Is there something that you're looking for? Mark Newman specifically called out the Analyst Day for Sandisk. Is there something you're looking for other than just continued macro that will drive this stock down towards your target? I'm looking for a catalyst for Kioxia. I mean, I know that's a long-term rating, you said, Mark Li, but I mean, is there something specific we should be looking for? I know with shorts, catalysts are very important.

ML: Yes. Anything on LTA could be a readacross for Kioxia, too. Even Kioxia may be less active in LTA discussion. But any update on that will be important. Anything on, say, pricing, near-term pricing will be also important. Not just for Kioxia, also for Samsung, SK, and Micron too.

Q: Got it. Can I ask you, maybe the Sandisk Analyst Day on the 13th could be a catalyst for Kioxia too, in a different way perhaps. Can I ask you both about SK Hynix and the US listing? Obviously, Mark, you're following the stock a little more closely. Fund flows certainly impact for Mark Newman's names as well. What are you anticipating with the ADR and anything specifically about how these stocks will trade over the next few weeks that we should be concerned or aware about?

ML: I think over time, the premium that Micron enjoyed for years should narrow if not reduce to zero with their ADRs. Then, yes, then certainly all sector, including ADRs too, will be subject to same structural or cyclical consideration. But in terms of multiple, yeah, I think SK will enjoy a bit of multiple expansion for a while. I don't know how long, but for a while. So the gap between the two, between SK and Micron should close.

Q: Mark Newman, any thoughts?

MN: Yeah, I mean, just to add, I think, investors often ask me about the multiple difference between Sandisk and the others — Micron, Hynix, Samsung, etc. And I think part of the reason is that the U.S. names trade higher, I think U.S. names in general trade higher, and so I think having a U.S. listing for Hynix is positive for them. But for the U.S. names, potentially a little bit of a short-term headwind. You're gonna have more investors who are going to be a little bit more focused on Hynix perhaps than they were in the past, for investors that are only investing in U.S. listings, of which there are still quite a lot. And so, potentially a bit of a headwind to Sandisk in the near term, but I don't think that's a long-term headwind.

Q: Fantastic. Well, listen, we are on the hour here. I just want to thank both of you. This was fantastic. I know I learned a lot. I hope, we can do this again, and, look forward to you two publishing jointly on some topics. It looks like you agree with each other a lot more than you disagree, but it's fun to identify where those differences of opinion lie. So, well done. Thank you both, and thank you all for joining us. Have a great day, everybody. If you have any questions that remain unanswered, please reach out to

your salesperson or directly to Mark Newman or Mark Li. Have a great day.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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VALUATION METHODOLOGY**SanDisk Corp**

We value SNDK at \$3000 on 11x FY28 EPS, or 14x through cycle EPS (FY26-30 avg EPS).

Samsung Electronics Co Ltd

We value Samsung Electronics with 6.2x of forward 5Q-8Q EPS of KRW71,172 and arrive at price target of KRW 440,000 for its common shares, KRW374,000 for preferred shares, and US\$7,350 for GDR.

SK Hynix Inc

We value SK hynix with 6.2x of forward 5Q-8Q EPS of KRW536,158 and arrive at price target of KRW3,300,000.

Micron Technology Inc

We value Micron with 7.7x of forward 5Q-8Q EPS of US\$169.8 and arrive at price target of US\$1,300.

KIOXIA Holdings Corp

We value KIOXIA with 4.1x of 1-year forward EPS of ¥9,810 and arrive at price target of ¥40,000.

RISKS**SanDisk Corp**

The biggest risks to the downside on Sandisk and to our price target are that: 1) Near-term numbers look high and face cyclical downside from NAND; 2) SNDK's disclosures and broader investor communications have been confusing, potentially dissuading more quality-focused investors; and 3) NAND weakness could extend beyond the current cycle and be more structural, in which case Sandisk's DCF value could be structurally lower and the impaired value of its assets may be materially below replacement cost.

Samsung Electronics Co Ltd

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

SK Hynix Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

Micron Technology Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

KIOXIA Holdings Corp

Upside risk to our target price include 1) stronger than expected NAND demand growth, which likely would lead to better NAND pricing and hence company earnings, 2) favorable policy support from Japanese government such as capex subsidies, and 3) better than expected cost reduction

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

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Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

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- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

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Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

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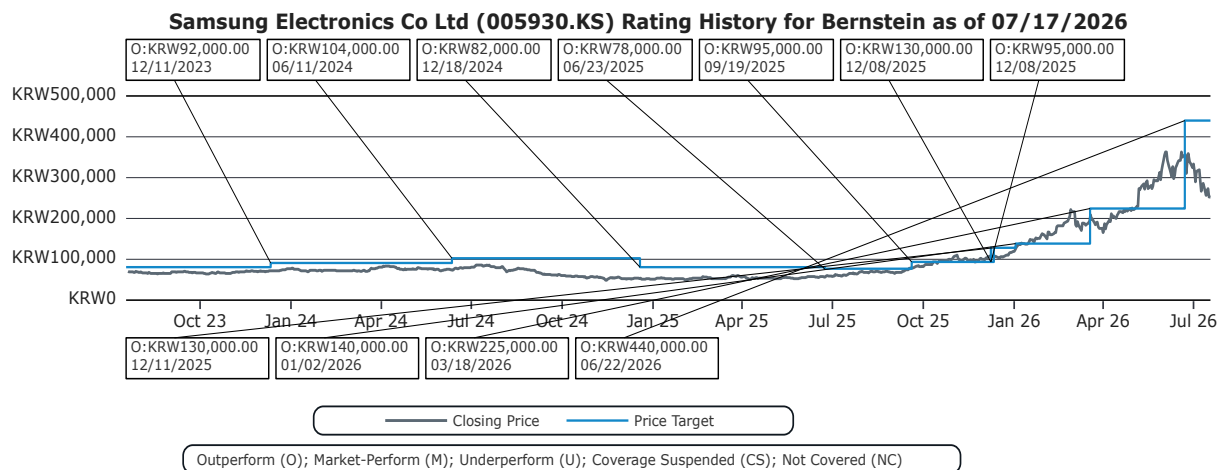
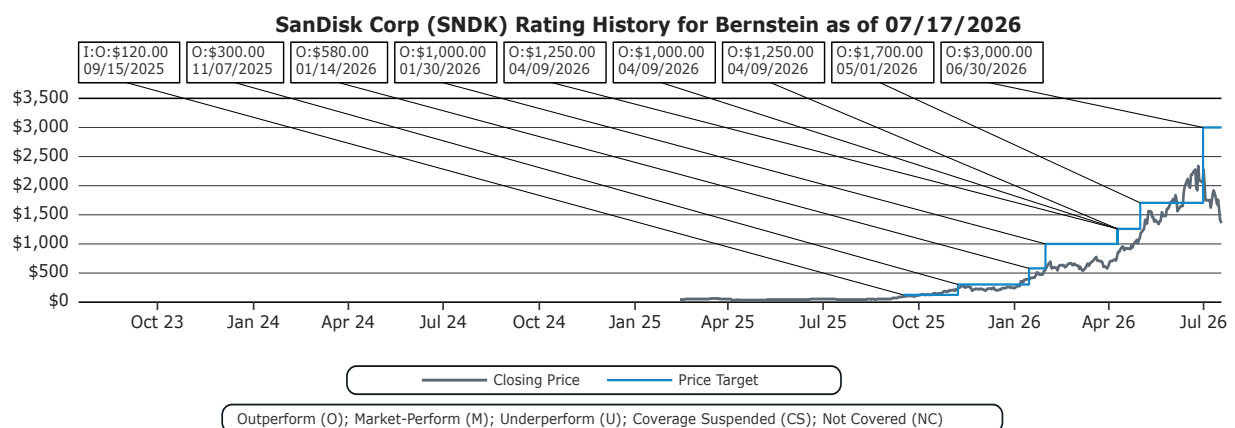
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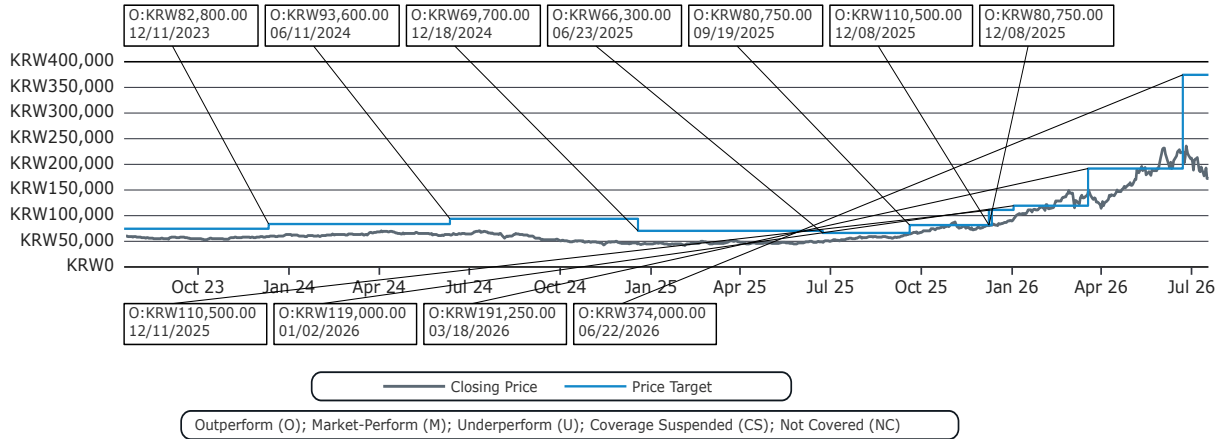
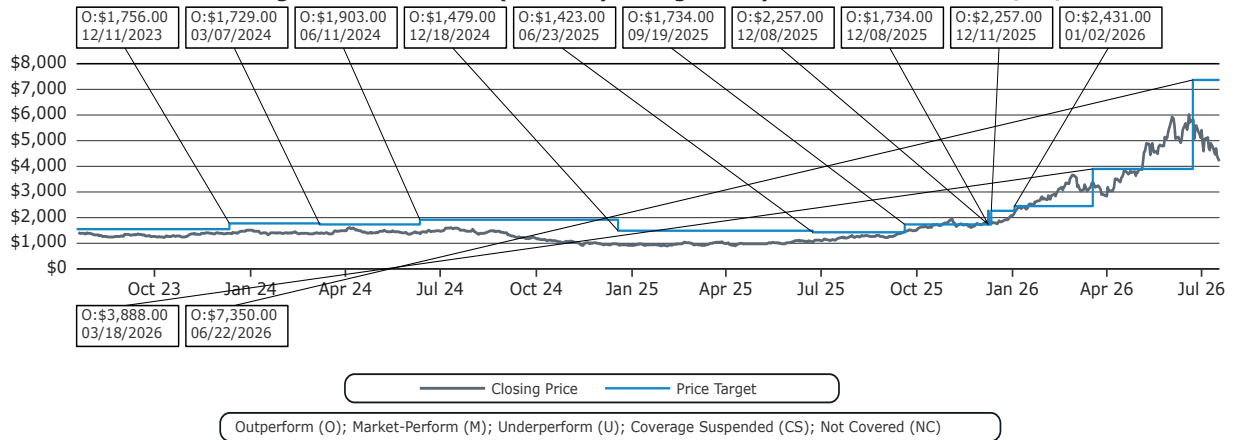
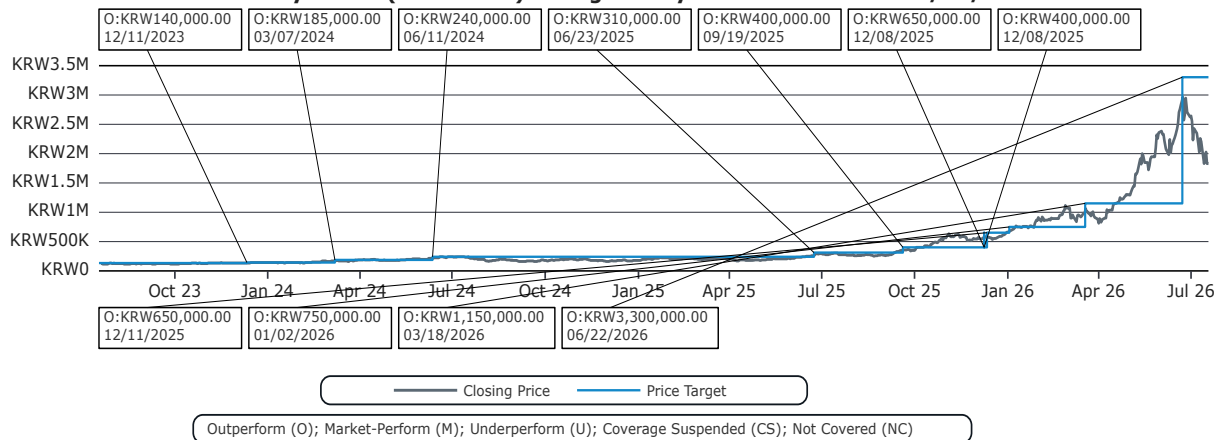
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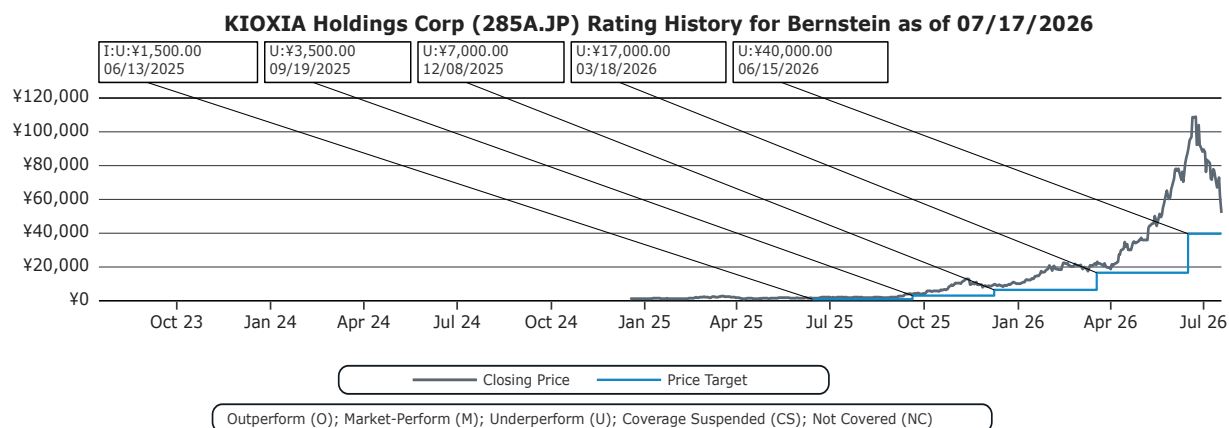
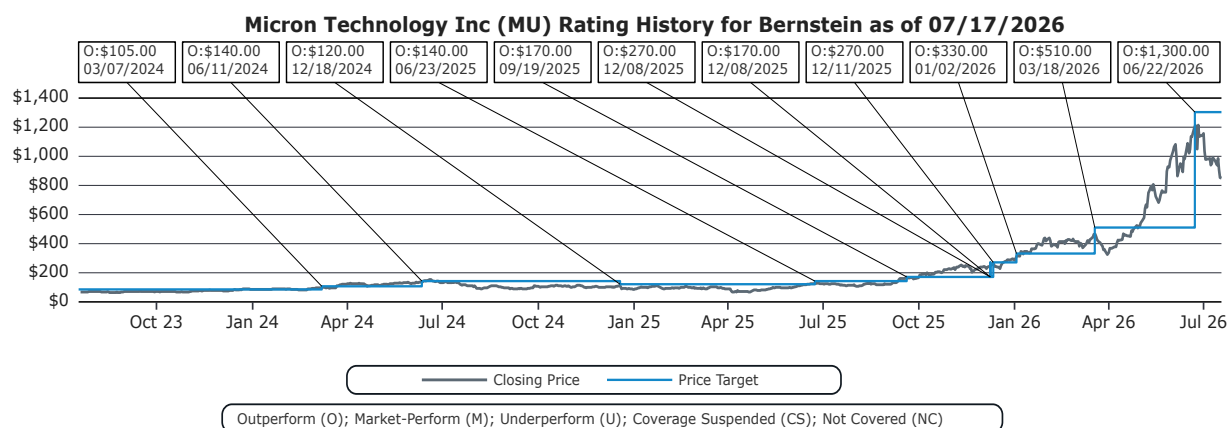
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Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY

Samsung Electronics Co Ltd (005935.KS) Rating History for Bernstein as of 07/17/2026**Samsung Electronics Co Ltd (SMSN.LI) Rating History for Bernstein as of 07/17/2026****SK Hynix Inc (000660.KS) Rating History for Bernstein as of 07/17/2026**



All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

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